Extended Kalman Filtering for State Estimation in Fed-Batch Bioreactor Systems Under Varying Noise Regimes

Zachary Hatzenbeller
EN.625.714 – Stochastic Differential Equations

Abstract

This project applies Extended Kalman Filter (EKF) methods to state estimation in a simulated 14-day fed-batch bioreactor running under Monod growth kinetics. The system has three states (viable cell concentration X_v, substrate glucose S, and monoclonal antibody titer P) and is observed through a mix of online sensors and sparse offline assays. I implemented the EKF from the discrete-time stochastic state-space formulation, which ties back to the Kalman–Bucy filter and Itō calculus covered in the course. A sweep over different process noise levels shows a bias–variance tradeoff, where the best performance on X_v and S is near the baseline Q setting. Whiteness tests on the innovations sequence (Ljung–Box and ACF) confirm the martingale property holds under correct model specification. Baseline RMSE values came out to 0.172, 0.171, and 0.055 for X_v, S, and P, respectively.

1. Introduction

One of the core problems in stochastic systems is figuring out the true state of a system when your observations are noisy and incomplete. The Kalman filter gives the minimum-variance unbiased estimate for linear Gaussian systems, and the Extended Kalman Filter (EKF) extends that idea to nonlinear systems by linearizing around the current state estimate at each step.

Fed-batch bioreactors are a good application for this because several things make them challenging to monitor directly. The growth dynamics are nonlinear (Monod kinetics), not all states can be measured continuously, sensors have different noise levels and sampling rates, and some measurements (like titer) only come from lab assays done once a day. Getting a good real-time estimate of cell concentration and titer matters a lot in pharmaceutical manufacturing for process control and meeting regulatory requirements.

The main question I wanted to explore is: how does the level and structure of process and measurement noise affect how well the EKF can track the true state? And can we use the innovations sequence as a tool to check whether the filter is working correctly? Section 2 covers the bioreactor model and the stochastic formulation. Section 3 walks through the EKF implementation. Section 4 describes the simulation setup. Section 5 presents the results, and Sections 6 and 7 discuss findings and wrap up.

2. Mathematical Formulation

2.1 Continuous-Time Process Model

The bioreactor state is $\mathbf{x}(t) = [X_v(t), S(t), P(t)]^T$, where X_v is viable cell concentration (10^6 cells/mL), S is glucose (g/L), and P is product titer (g/L). The dynamics follow a standard fed-batch ODE model:

$$dX_v/dt = \mu(S)\, X_v - k_d\, X_v$$
$$dS/dt = -\mu(S)\, X_v \,/\, Y_{x/s} - m_s\, X_v + F_s(t) \,/\, V(t)$$
$$dP/dt = q_p\, X_v$$

where the specific growth rate follows Monod kinetics: $\mu(S) = \mu_{max} S / (K_s + S)$. The parameter values used are: $\mu_{max} = 0.035\ h^{-1}$, $K_s = 0.5$ g/L, $k_d = 0.003\ h^{-1}$, $Y_{x/s} = 2.5$, $m_s = 0.005\ g{\cdot}g^{-1}{\cdot}h^{-1}$, and $q_p = 0.01\ g{\cdot}cell^{-1}{\cdot}h^{-1}$. The feed $F_s(t)$ is a Gaussian-smoothed bolus every 24 hours starting at t = 48 h, adding 5% volume at 400 g/L each time.

2.2 Stochastic Discrete-Time Representation

To apply the Kalman filter, I discretized the system and added noise terms to get the standard stochastic state-space form:

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k, u_k) + \mathbf{w}_k,\ \mathbf{w}_k \sim N(\mathbf{0}, Q)$$
$$\mathbf{z}_k = h(\mathbf{x}_k) + \mathbf{v}_k,\ \mathbf{v}_k \sim N(\mathbf{0}, R)$$

The function $f(\cdot)$ is the nonlinear state transition computed using RK4 over $\Delta t = 0.5$ h, and $h(\cdot)$ maps states to measurements. The process noise Q is meant to capture model uncertainty. This setup is the discrete-time analog of the Kalman–Bucy filter: in the continuous limit, Q approaches $G\Sigma G^T\Delta t$ where Σ is the diffusion coefficient of the underlying Wiener process.

2.3 Measurement Model and Sensor Fusion

The filter fuses two types of measurements. Online sensors run every 0.5 h: capacitance ($z_{cap} = \alpha X_v$ + noise, $\alpha = 1.5$, $\sigma_{cap} = 0.6$) and Raman spectroscopy ($z_{ram} = \beta S$ + noise, $\beta = 0.8$, $\sigma_{ram} = 0.5$). Offline assays of X_v, S, and P are available every 24 h with lower noise ($\sigma_{VCC} = 0.15$, $\sigma_{gluc} = 0.20$, $\sigma_{titer} = 0.10$). When both types are available at the same time step, I stack them into a 5x1 vector with a block-diagonal R so the filter processes them together.

2.4 Martingale Structure of the Innovations

The innovations at each step are $\nu_k = z_k - h(\hat{x}_{k|k-1})$. When the filter model is correct, ν_k should form a martingale difference sequence: $E[\nu_k \mid F_{k-1}] = 0$. This means the innovations carry no predictable information, only unpredictable noise. If they are autocorrelated or have a nonzero mean, something is wrong with the model or noise settings. The normalized innovations $\bar{\nu}_k = S_k^{-1/2}\ \nu_k$ should look like i.i.d. N(0, I) if things are working correctly, which we can check with whiteness tests.

3. Extended Kalman Filter Implementation

3.1 Predict-Update Recursion

Since the bioreactor model is nonlinear, I used the EKF which linearizes $f(\cdot)$ and $h(\cdot)$ at each time step. The predict step uses RK4 for the state and the Euler-discretized Jacobian for the covariance:

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1|k-1}, u_{k-1})\ \text{[RK4]}$$
$$P_{k|k-1} = F_{k-1}\ P_{k-1|k-1}\ F_{k-1}^T + Q$$

For the update, I compute the Kalman gain $K_k = P_{k|k-1}\ H_k^T\ S_k^{-1}$ and then update both state and covariance. I used the Joseph form for the covariance update to keep it numerically stable: $P_{k|k} = (I - K_k H_k)\ P_{k|k-1}\ (I - K_k H_k)^T + K_k R K_k^T$. This guarantees the covariance stays positive semi-definite even after many iterations.

3.2 Analytical Jacobian for Monod Kinetics

Rather than using numerical differentiation, I worked out the analytical Jacobian of the continuous-time system. The tricky part is the Monod term, which gives $\partial\mu/\partial S = \mu_{max} K_s / (K_s + S)^2$. The Jacobian rows are:

$$J[1,1] = \mu - k_d,\ J[1,2] = (\partial\mu/\partial S)\ X_v,\ J[1,3] = 0$$
$$J[2,1] = -\mu/Y_{x/s} - m_s,\ J[2,2] = -(\partial\mu/\partial S)\ X_v / Y_{x/s},\ J[2,3] = 0$$
$$J[3,1] = q_p,\ J[3,2] = 0,\ J[3,3] = 0$$

The state-transition Jacobian is then $F = I + \Delta t\, J$ (first-order Euler). One thing worth noting is that the off-diagonal coupling between X_v and S means estimation errors in one state bleed into the other through the Monod term.

3.3 Mixed-Frequency Measurement Fusion

At each time step, the filter checks which measurements are available. If only online measurements are there, it uses a 2x1 measurement vector. If only offline assays are available, it uses a 3x1 vector. If both coincide (every 24 hours), it fuses all five measurements in one update step. This naturally handles the different sampling rates without any interpolation.

4. Simulation Setup

I generated the ground truth by integrating the ODE system using SciPy's RK45 solver with tight tolerances (rtol $= 10^{-8}$, atol $= 10^{-10}$) over 336 hours (14 days), giving 673 time points at $\Delta t = 0.5$ h. Initial conditions were $X_v(0) = 0.5$, $S(0) = 6.0$ g/L, and $P(0) = 0$ g/L.

To test convergence, I started the filter with a 20% low initial guess on X_v and a 20% high guess on S, with initial covariance $P_0 = \mathrm{diag}(0.1, 1.0, 0.01)$. Measurements were generated by adding Gaussian noise to the true trajectory at the appropriate sampling rates (fixed random seed throughout). The baseline process noise was $Q = \mathrm{diag}(0.001, 0.01, 0.0001)$.

5. Results

5.1 Baseline EKF Performance

Figure 1 shows the EKF estimates alongside the true trajectory and measurements for all three states. The filter recovers from its initial offset fairly quickly and stays close to the true values for the full 14 days.

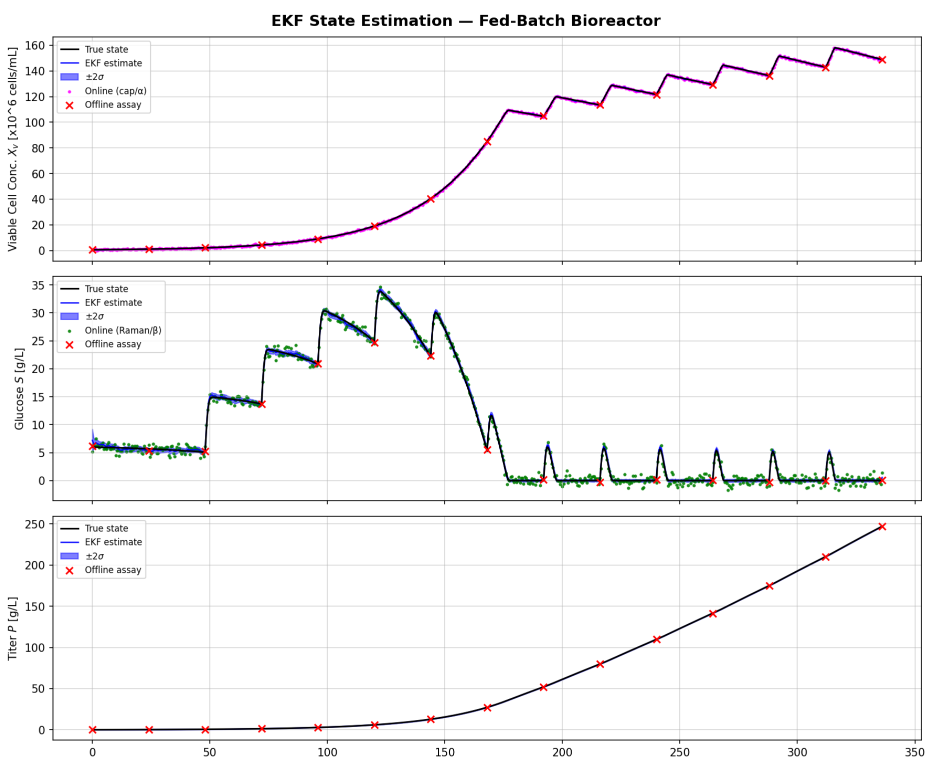

Figure 1. EKF state estimation over the 14-day fed-batch run. Top: viable cell concentration with capacitance online measurements (divided by α) and offline VCC assays. Middle: glucose with Raman measurements (divided by β) and offline

assays; the feed boluses show up as step increases. Bottom: titer with offline assays. Blue shading is the ±2σ confidence region. RMSE: X_v = 0.172, S = 0.171, P = 0.055.

The glucose panel is the most interesting to look at because you can clearly see the filter responding to each feed bolus. The state jumps up, the filter initially has wider uncertainty, and then tightens back up as it assimilates more online data. The titer estimate is the cleanest of the three because it only increases over time and gets anchored every 24 hours by the offline assay. The RMSE on titer (0.055) is much lower than X_v and S for that reason.

5.2 Innovations Whiteness Test

Figure 2 shows the normalized innovations and their autocorrelation functions for both online channels. This is the main diagnostic for whether the filter is working correctly.

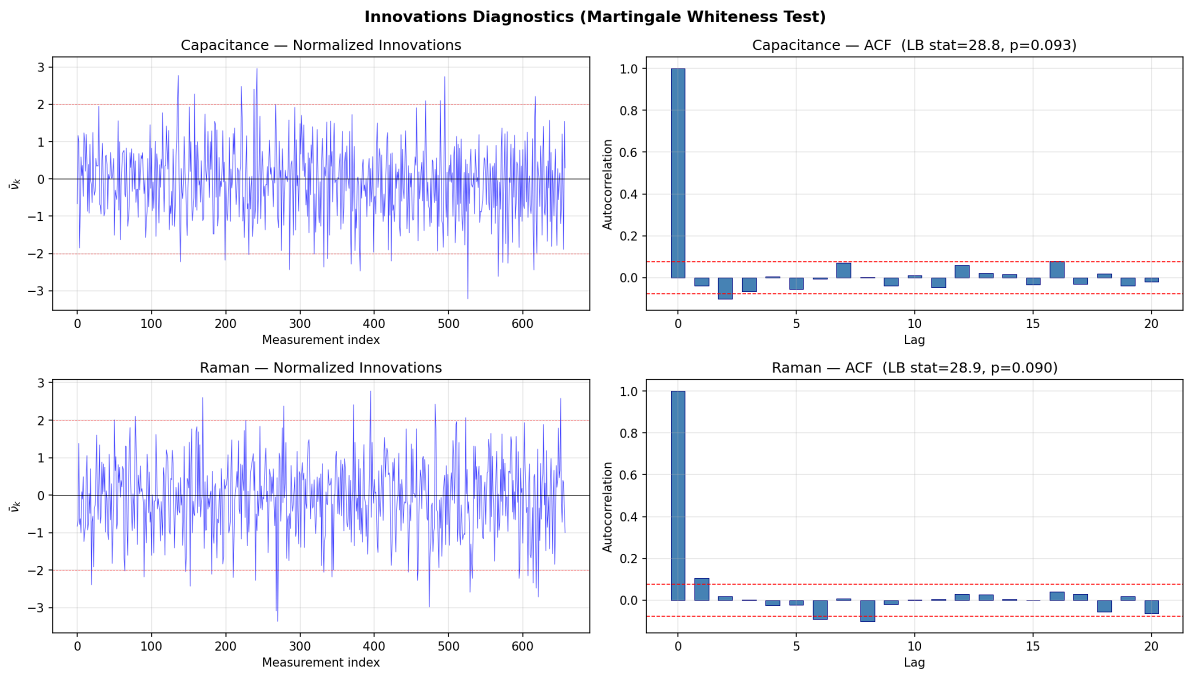

Figure 2. Innovations diagnostics. Left: normalized innovations time series for the capacitance (top) and Raman (bottom) channels. Right: sample ACF with 95% white-noise confidence bounds (dashed red). Ljung–Box statistics: capacitance LB = 28.8 (p = 0.093), Raman LB = 28.9 (p = 0.090).

The normalized innovations look roughly zero-mean and stay within the ±2 range most of the time, which is what you would expect from a standard normal. The ACF plots show almost all lags are within the 95% confidence bands ($\pm 1.96/\sqrt{n}$), meaning there is no significant autocorrelation. The Ljung–Box test at lag 20 gives p = 0.093 for capacitance and p = 0.090 for Raman. Both are above 0.05, so we do not reject the null hypothesis that the innovations are white. This confirms the martingale property holds for the baseline filter setup.

5.3 Noise Regime Analysis

Figure 3 shows how RMSE changes as I scale the process noise Q from 0.1 to 10 times the baseline, keeping R fixed.

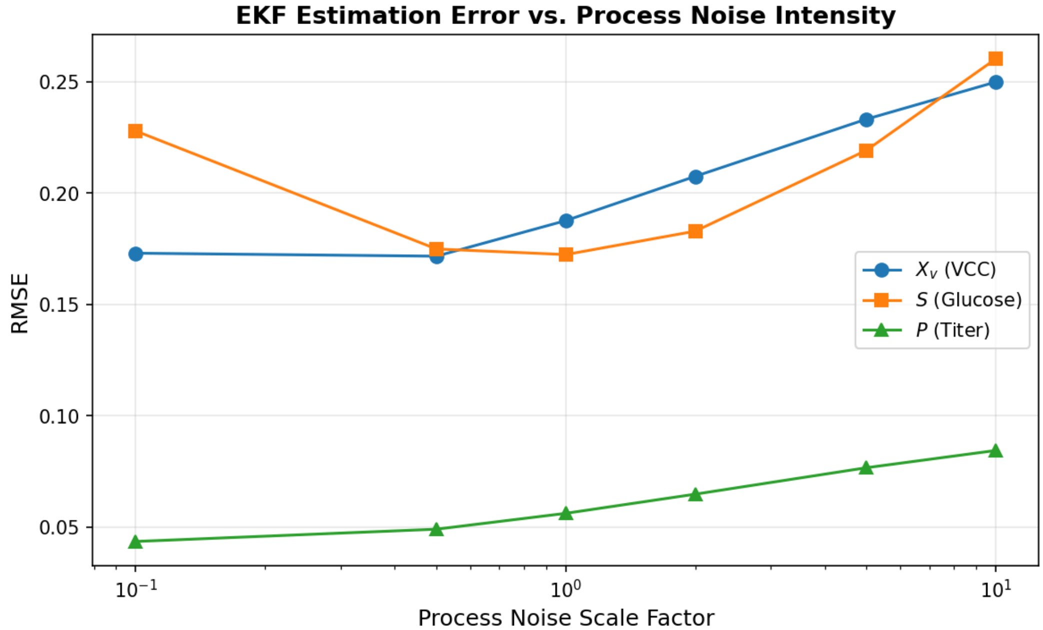

Figure 3. EKF RMSE versus process noise scale factor (log scale) for viable cell concentration (X_v), glucose (S), and titer (P). Each point is a full 14-day simulation using the same synthetic measurements.

There is a pretty clear tradeoff visible in the results. For X_v and S, RMSE stays fairly flat at low Q values, hits a minimum somewhere around scale factor 0.5 to 1.0, then climbs again at higher values. When Q is too small, the filter trusts the ODE model too much and reacts slowly to feed disturbances. When Q is too large, it starts chasing the noisy measurements and the estimates get noisier. Titer is a different story: RMSE increases monotonically with Q-scale, which makes sense because P only shows up through $dP/dt = q_p X_v$, so any extra noise in the X_v estimate gets integrated directly into the titer error.

The fact that the baseline Q (scale factor 1.0) sits near the minimum for X_v and S suggests it was set reasonably well to begin with.

6. Discussion

6.1 Connection to SDE Theory

The EKF implemented here is the discrete-time approximation of the Kalman–Bucy filter, which is the continuous-time solution to optimal filtering for SDE systems. In continuous time the system looks like $d\mathbf{x} = f(\mathbf{x}, t)dt + G\, dW_t$ with observations $d\mathbf{z} = h(\mathbf{x})dt + dV_t$, and the optimal filter has to solve the Zakai equation for the conditional density. For Gaussian systems this simplifies down to ODEs for the mean and covariance, which is the Kalman–Bucy filter. The EKF gives you the same thing in discrete time using local linearizations.

The innovations ν_k are the discrete-time version of the Itō integral $d\iota_t = dz_t - h(\hat{x}_t)dt$, which appears in the Kalman–Bucy gain term. Checking that $E[d\iota_t \mid F_t^-] = 0$ in the discrete case via the Ljung–Box test is the same idea, just applied to the sample innovations sequence.

6.2 Observability and Sensor Design

With both online and offline sensors available, all three states are observable. Titer P is the one that benefits most from offline assays, since the only way to directly correct the titer estimate is from the lab measurement every 24 hours. Between assays, the filter relies on integrating $q_p X_v$, which means errors in X_v accumulate into the titer estimate over time. The offline anchors help reset this drift.

The Monod coupling in the Jacobian means X_v and S estimation errors are linked. This cross-coupling is probably why the optimal Q-scale is slightly different for X_v and S in the noise sweep.

6.3 Limitations

A few things could be improved. The Euler Jacobian discretization ($F = I + \Delta t\ J$) is only first-order accurate and could accumulate error over the 14-day run. The filter also assumes Q and R are known exactly, which would not be the case in a real bioreactor where you would need some form of adaptive noise estimation. And the feed profile $F_s(t)$ is treated as perfectly known here, which is an idealization since pump errors and feed concentration uncertainty would also need to be accounted for in practice.

7. Conclusion

This project walked through applying the EKF to a simulated fed-batch bioreactor and connecting it back to the stochastic process theory covered in the course. A few key takeaways:

- The EKF with RK4 propagation and the analytical Monod Jacobian tracked all three states reasonably well, with baseline RMSE of 0.172 (X_v), 0.171 (S), and 0.055 (P), even starting from a biased initial condition.
- The innovations whiteness test ($p \approx 0.09$ for both channels) confirmed the martingale property holds for the baseline filter, which connects directly to the theory of optimal filtering and the innovations representation from class.
- The noise sweep showed that estimation accuracy is sensitive to how Q is set. Too small and the filter is too rigid; too large and it gets noisy. Titer is especially sensitive because errors in X_v integrate into P over time.
- Using the Joseph form and handling online and offline measurements separately at each time step kept the filter numerically stable throughout the full 14-day run.

Overall, this was a good exercise in seeing how the abstract ideas from the course (the Kalman–Bucy filter, martingale innovations, Itō calculus) translate into something practical. The bioreactor setting made it concrete and showed that the filter diagnostics actually give you useful information about whether your noise model is tuned correctly.

References

[1] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

[2] A. H. Jazwinski, *Stochastic Processes and Filtering Theory*. Academic Press, 1970.

[3] D. Simon, *Optimal State Estimation: Kalman, H∞, and Nonlinear Approaches*. Wiley-Interscience, 2006.

[4] S. Särkkä, *Bayesian Filtering and Smoothing*. Cambridge University Press, 2013.

[5] J. Monod, "The growth of bacterial cultures," *Annual Review of Microbiology*, vol. 3, no. 1, pp. 371–394, 1949.

[6] G. Bastin and D. Dochain, *On-line Estimation and Adaptive Control of Bioreactors*. Elsevier, 1990.

[7] B. Øksendal, *Stochastic Differential Equations: An Introduction with Applications*, 6th ed. Springer, 2003.

[8] G. M. Ljung and G. E. P. Box, "On a measure of lack of fit in time series models," *Biometrika*, vol. 65, no. 2, pp. 297–303, 1978.

[9] A. Gelb (Ed.), *Applied Optimal Estimation*. MIT Press, 1974.

[10] F. J. Doyle et al., "Model identification of signal transduction networks from cell culture assay data," *Metabolic Engineering*, vol. 4, no. 1, pp. 64–76, 2002.